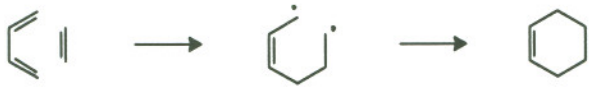
Organic Chemistry Cumulative Examination
November 12, 1994

1. *J. Am. Chem. Soc.* **1994**, *116*, 9675

There is *still* a great deal of debate about the mechanism of the Diels–Alder reaction, as to whether the two new bonds are formed at the same time (concerted mechanism), or one by one (diradical mechanism).

concerted
transition state



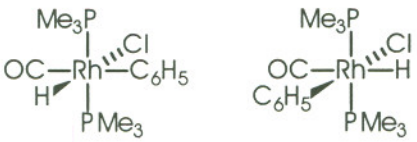
diradical intermediate
(note allyl radical structure)

The authors of the paper cited above suggest using secondary deuterium kinetic isotope effects as a possible way of deciding between the two mechanisms. There are ten hydrogens in the unsubstituted system. Predict whether a normal or inverse kinetic isotope effect will be observed for deuterium substitution at each hydrogen, if the concerted mechanism is operational. Do the same for the diradical mechanism, assuming the first step is rate determining.

Helpful information: (1) Normal isotope effect means the reaction with hydrogen is faster than the reaction with deuterium. Inverse isotope effect means the reaction with deuterium is faster than the reaction with hydrogen. (2) Increased substitution at a given atom (such as going from an sp2 to sp3 hybridization) causes an increased resistance to bending motion (increased bending force constant). (3) The bending force constant at a radical center is less than that of an alkene.

2. *J. Am. Chem. Soc.* **1994**, *116*, 9492

Reaction of the square planar rhodium complex, *trans*-$Rh(PMe_3)_2(CO)Cl$ with benzene resulted in a variety of products, among which are the two isomeric octahedral complexes shown below.



How would you distinguish between these two using NMR spectroscopy? Assume that your spectrometer has ^{1}H, ^{13}C, ^{31}P capabilities, and you can do any ^{13}C enrichment you want. You can also assume that all the NMR peaks have been assigned.

3. *J. Am. Chem. Soc.* **1994**, *116*, 9471

Propose a mechanism for the following <u>photochemical</u> reaction.

Hint: the mechanism I have in mind has five steps.